

PDF Merger and Splitter Project Report

1. Introduction

1.1 Project Overview

This "PDF Merger and Splitter" project is a Python-based tool designed to enable users to merge multiple PDF files into a single file or split a PDF into separate single-page files. It utilizes the PyPDF2 library to demonstrate practical application of programming concepts related to file handling and document management.

1.2 Problem Statement

Users often need to combine several PDF documents or extract specific pages from large PDFs for easier sharing or editing. Manual handling can be time-consuming, error-prone, or require expensive software.

1.3 Scope of the Project

- Merge any number of PDFs into one file.
- Split any PDF into multiple one-page PDFs.
- Run through a simple command-line interface.
- Limitations: Does not support encrypted or password-protected PDFs.

2. System Requirements

2.1 Hardware Requirements

- Standard computer with a minimum of 2GB RAM.

2.2 Software Requirements

- Python 3.x
- PyPDF2 library (install with pip)
- Any OS supporting Python (Windows, macOS, Linux)

3. System Design and Methodology

3.1 Development Methodology

The project follows an iterative development approach with coding, testing, and refinement phases.

3.2 System Architecture

- Command-line interface for user choices.
- Separate modules/functions for merging and splitting PDFs.
- File validation and error handling included.

3.3 Data Flow

- Input: User provides file names and options via CLI.
- Process: Script reads PDFs, merges or splits using PyPDF2.
- Output: Generates new PDF(s) saved in specified location.

Testing and Results

4.1 Testing Strategy

Unit tests on functions for merging and splitting different PDFs with various page counts.

4.2 Test Cases

Test ID	Description	Input Files	Expected Output	Result
TC001	Merge 3 PDFs	file1.pdf, file2.pdf, file3.pdf	Single PDF with all pages	Pass
TC002	Split 1 PDF	bigfile.pdf	Multiple single-page PDFs in folder	Pass

4.3 Outcome

The tool successfully merges multiple PDFs and splits PDFs into single pages with no errors.

5. Conclusion and Future Scope

5.1 Conclusion

The project met all objectives by providing a functional PDF merger and splitter tool implemented in Python.

5.2 Future Enhancements

- Add support for encrypted PDF files.
- Develop a graphical user interface (GUI) for ease of use.
- Include options to merge/split specific page ranges.

